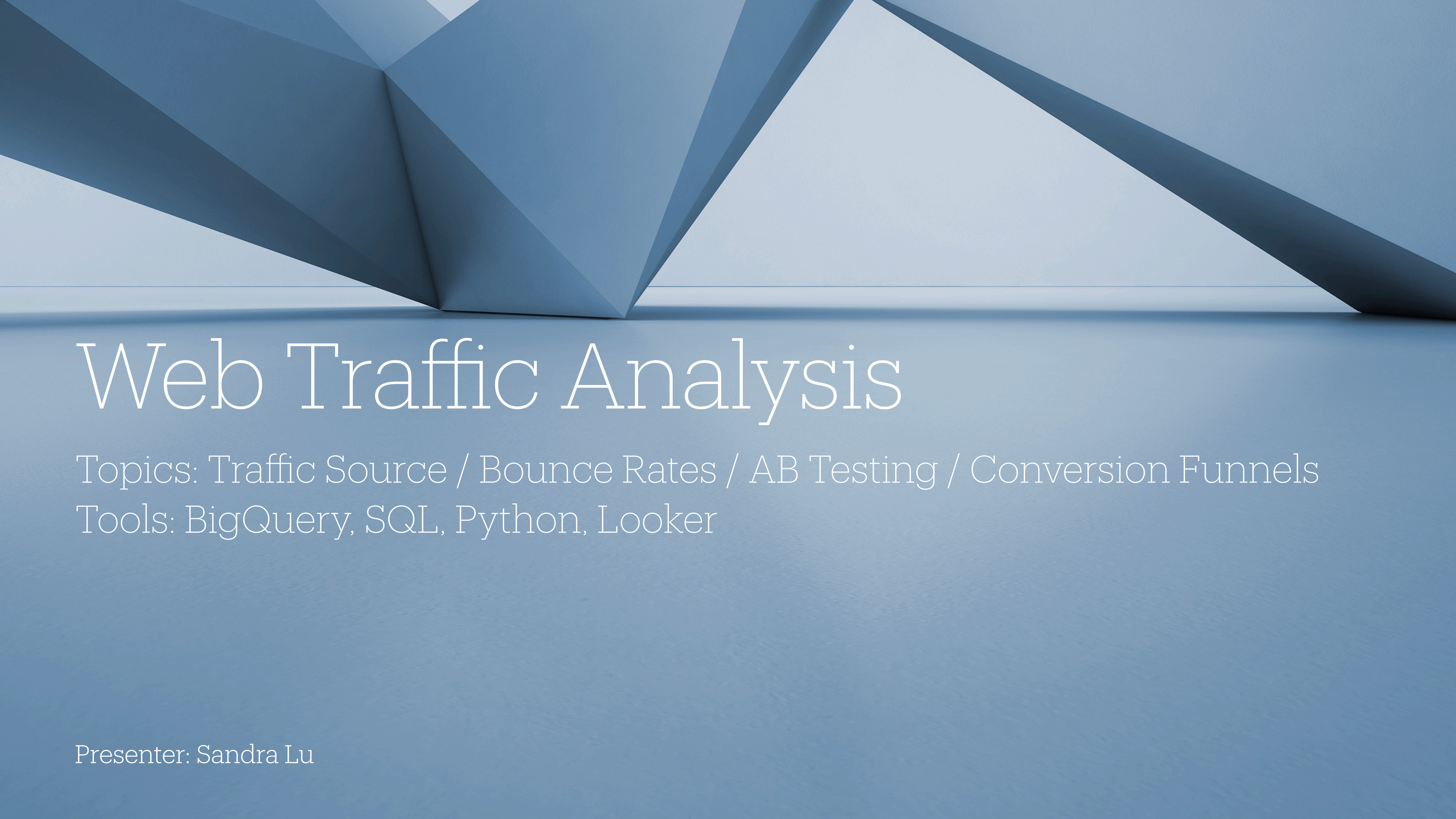




Web Traffic Analysis

Topics: Traffic Source / Bounce Rates / AB Testing / Conversion Funnels

Tools: BigQuery, SQL, Python, Looker

Presenter: Sandra Lu

Background

About the Company

Maven Fuzzy Factory is a newly launched frictional eCommerce retail startup that sells stuffed animals. The company was started on march 2012 and has been in operation for 3 years now.



Background

About the Dataset

This project utilized a custom-built e-commerce database for Maven Fuzzy Factory. The database contains six related tables with eCommerce data about:

- Website Sessions
- Website Pageviews
- Products
- Orders
- Order Items
- Order Item Refunds

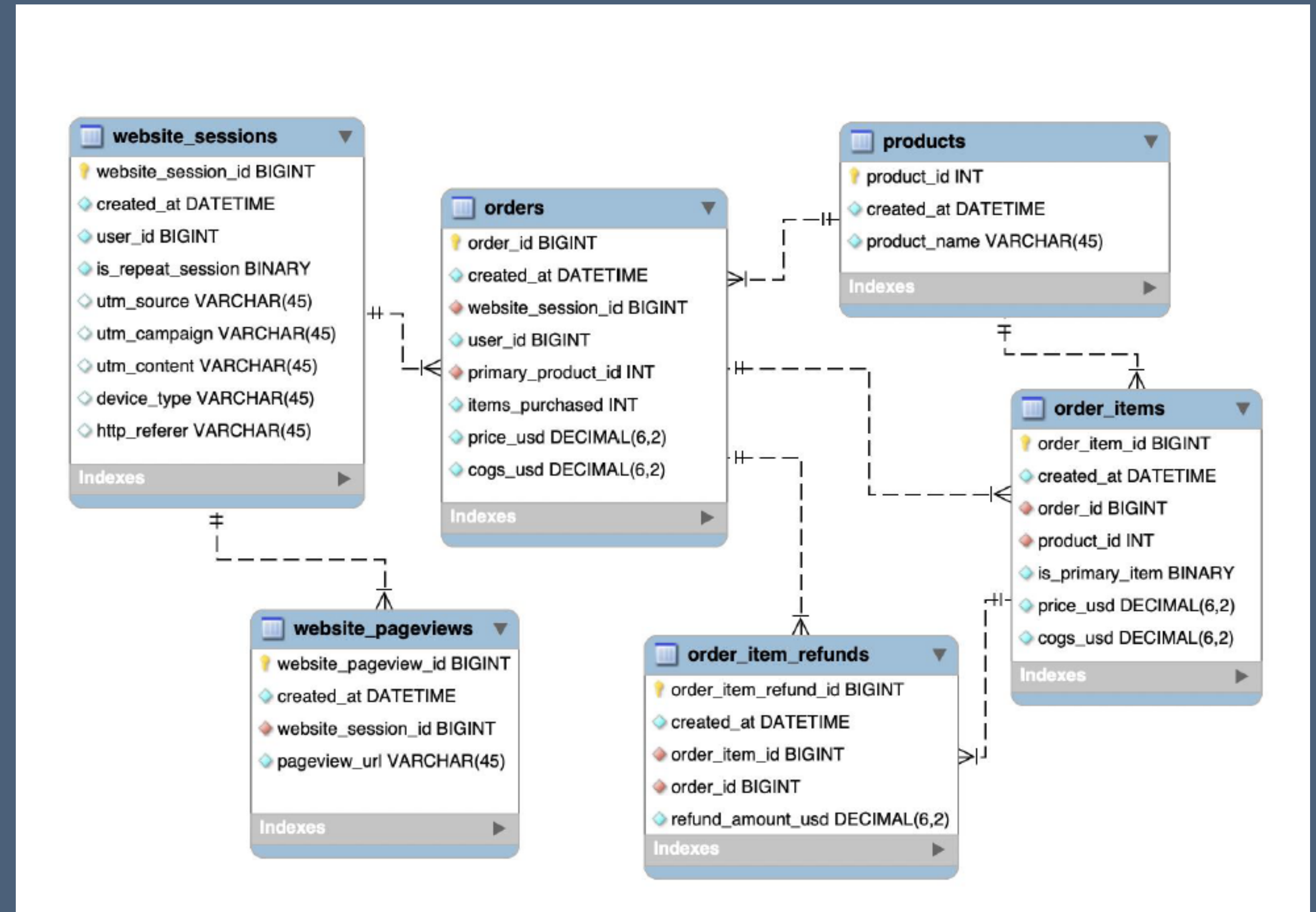


Figure 1: Entity Relationship Diagram (ERD) for Six Tables

Background

About channels

Row	utm_source ▼	utm_campaign ▼	http_referer ▼
1	NULL	NULL	NULL
2	NULL	NULL	https://www.bsearch.com
3	NULL	NULL	https://www.gsearch.com
4	bsearch	brand	https://www.bsearch.com
5	bsearch	nonbrand	https://www.bsearch.com
6	gsearch	brand	https://www.gsearch.com
7	gsearch	nonbrand	https://www.gsearch.com
8	socialbook	desktop_targeted	https://www.socialbook.com
9	socialbook	pilot	https://www.socialbook.com

Direct Type-In

Organic Search

Paid Search

Figure 2: List of UTM sources, campaigns, and referral domain (i.e. http_referer)

Project Objective

Part 1 - Traffic Source Analysis

- Identifying the most useful traffic channels, campaigns, and keywords with strongest conversation rates to make financial decisions and shift budget towards most effective traffic sources.
- Analyzing user behavior and patterns in different traffic sources to generate strategy and improve user experience.
- Comparing the most to least effective traffic to identify and eliminate wasteful traffic sources and scale the most effective source.

Project Objective

Part 2 - Website performance and analysis

- Identify the most and least viewed website pages by the customers to make creative decisions on the enhancement of the pages.
- Analyzing the conversation funnels of customers to identify the most common path customers, take before purchasing products (from landing page to sale) and lower bounce rate.
- Understanding the pattern and effect of website pages on customer orders and make changes to the website pages and push maximum products to customer orders.

Project Objective

Part 3 - Business patterns and analyzing seasonality

- Analyzing seasonality to better understand the pattern, spikes, or slowdowns in demand.
- Analyzing sales and revenue and identify most busy time periods to make strategic business decisions.
- Watching product sales trend to understand the overall health of the business

Section 1 - Traffic Source Analysis

Which Channels drive traffic to the website?

Row	utm_source ▼	utm_campaign ▼	http_referer ▼	sessions ▼	% of total ▼
1	gsearch	nonbrand	https://www.gsearch.com	282706	59.79
2	bsearch	nonbrand	https://www.bsearch.com	54909	11.61
3	NULL	NULL	NULL	39917	8.44
4	NULL	NULL	https://www.gsearch.com	35202	7.44
5	gsearch	brand	https://www.gsearch.com	33329	7.05
6	NULL	NULL	https://www.bsearch.com	8209	1.74
7	bsearch	brand	https://www.bsearch.com	7914	1.67
8	socialbook	desktop_targeted	https://www.socialbook.com	5590	1.18
9	socialbook	pilot	https://www.socialbook.com	5095	1.08

Figure 3: Total Sessions and % of Total for UTM source, campaign, and http_referrer

- Google Search with non-brand campaign is the primary driver of our business. This is the major traffic source.
- Brand traffic has very low traffic sessions during 2012/03-2015/03, compared to non-brand traffic.

Monthly trend split out non-brand vs. brand campaigns

Row	year ▼	month ▼	nonbrand_sessions	brand_sessions ▼	nonbrand_orders ▼	brand_orders ▼
1	2012	3	1852	8	60	0
2	2012	4	3509	65	86	6
3	2012	5	3295	115	91	6
4	2012	6	3439	139	114	7
5	2012	7	3660	151	136	9
6	2012	8	4673	204	174	10
7	2012	9	4227	264	172	16
8	2012	10	5197	337	219	15
9	2012	11	8506	383	356	17

Figure 6: Monthly Trend for brand vs. non-brand sessions and orders

- Overall, the trend for brand sessions and brand orders is positive. Both metrics have been increasing steadily over the past 8 months, which suggests that the brand is becoming more well-known and trusted by consumers.
- But still the brand sessions are much lower than non-brand sessions. It was about 30:1 ratio for sessions and a little less than 15:1 ratio for orders.

What is the conversion rate (CVR) for those digital channels?

Row	utm_source	utm_campaign	sessions	orders	conversion_rate
1	bsearch	brand	7914	701	0.0886
2	gsearch	brand	33329	2511	0.0753
3	NULL	NULL	83328	6118	0.0734
4	bsearch	nonbrand	54909	3818	0.0695
5	gsearch	nonbrand	282706	18822	0.0666
6	socialbook	desktop_targeted	5590	288	0.0515
7	socialbook	pilot	5095	55	0.0108

Figure 4: Sessions, orders, and session-to-order conversion rate for UTM source and campaign

- Bing search engine with branded campaign has the highest session to order conversation rate (8.86%) followed by gsearch-brand and organic/direct search.
- Social books has the lowest conversion rate.
- Hence it might be a good idea to decrease the budget for those campaigns and allocate it to others.

What is the overall monthly conversion rate (CVR)?

Row	year ▼	month ▼	sessions ▼	orders ▼	conversion_rate ▼
1	2012	3	1860	60	0.0323
2	2012	4	3574	92	0.0257
3	2012	5	3410	97	0.0284
4	2012	6	3578	121	0.0338
5	2012	7	3811	145	0.038
6	2012	8	4877	184	0.0377
7	2012	9	4491	188	0.0419
8	2012	10	5534	234	0.0423
9	2012	11	8889	373	0.042

Figure 5: Monthly Trend of Sessions, Orders, and Session-to-Order conversion rate for Year 2012

- Overall, the business is seeing positive results in terms of both website traffic, orders and conversions.

Which type of device brings the most sessions through gsearch engine?

Row	device_type	sessions	orders	conversion_rate
1	mobile	10955	142	0.013
2	desktop	29069	1352	0.0465

Figure 7: Total Sessions, Orders, and Conversion Rate aggregated by Type of Device

- There is significant variation in conversion rates by device type, with desktop devices having an average conversion rate of 4.65% and mobile devices having a conversion rate of 1.3%. Desktop devices convert sessions to orders at a rate of 257.69% higher than mobile devices.
- This means that for every 100 mobile sessions which convert to orders, 257 more desktop sessions will convert in addition to taking mobile ones into account.
- Businesses may also want to invest more in desktop advertising since desktop sessions are more likely to convert to orders. So based on this analysis, I think we should plan to increase our bids for desktop advertising. By increasing our bids, we will achieve a higher ranking in search engine auctions, which I believe should lead to a sales boost.

Section 2 - Website performance and analysis

What is the most viewed website page, ranked by session volume?

Row	pageview_url	page_views
1	/home	11682
2	/products	4789
3	/the-original-mr-fuzzy	3418
4	/cart	1476
5	/shipping	979
6	/billing	807
7	/thank-you-for-your-order	349

Figure 8: Number of page views grouped by page view URL

Row	landing_page_url	sessions_hitting_this_landing_page
1	/home	11682

Figure 9: Number of Sessions hitting each landing page

- The homepage, the products page, and the Mr. Fuzzy page attract the majority of our website traffic.
- Looks like our CVR was low due to the fact that very less number of sessions billing page and hitting order page.
- Also, our traffic all comes in through the homepage right now. So that is the very first place where customer is seeing our website for the first time. Seems pretty obvious that this is the page where we should focus our efforts to improve the user experience.

What is the bounce rates for traffic landing on the homepage?

Row	landing_page_url	sessions	bounced_sessions	bounce_rate
1	/home	11682	6893	0.5901

Figure 10: Sessions, Bounced Sessions, and Bounce Rate on landing page

- The current homepage has a bounce rate of 60%, which is higher especially for high-quality traffic from paid search campaigns.
- I recommend that we create a custom landing page for paid search traffic. A custom landing page can be tailored to the specific needs of paid search traffic, and it can provide a more relevant and engaging experience for visitors. Then we should set up an experiment to compare the performance of the new page to the current homepage

The website manager ran a new custom landing page (/lander-1) in a 50/50 A/B testing against the homepage(/home) for gsearch nonbrand traffic from Jun 19 — July 28

Evaluate the new lander page performance by comparing bounce rates for the two groups.

Row	landing_page_url	sessions	bounced_sessions	bounce_rate
1	/home	2261	1319	0.5834
2	/lander-1	2316	1233	0.5324

Figure 11: Compare Session, Bounced Sessions, Bounce Rate for Landing Page Test

- From this analysis, we can see that the custom landing page has a lower bounce rate than the homepage, which is a great success.
- So it is no brainier that we now need to get campaigns updated so that all non-brand paid traffic is pointing to the new page.
- I think it is important to track and confirm if our traffic volume is all running to the new custom lander after campaign updates and also keep an eye on bounce rates trends.

Weekly trend of the volume of paid search non-brand traffic landing on /home and /lander-1.

Row	week_start_date	bounce_rate	home_sessions	lander_sessions
1	2012-06-01	0.6057	175	0
2	2012-06-03	0.5871	792	0
3	2012-06-10	0.616	875	0
4	2012-06-17	0.5582	492	350
5	2012-06-24	0.5828	369	386
6	2012-07-01	0.5821	392	388
7	2012-07-08	0.5668	390	411
8	2012-07-15	0.5424	429	421
9	2012-07-22	0.5138	402	394
10	2012-07-29	0.4971	33	995
11	2012-08-05	0.5382	0	1087
12	2012-08-12	0.514	0	998
13	2012-08-19	0.501	0	1012
14	2012-08-26	0.539	0	833

Before test, Entry page was home

Landing Page test is being conducted to judge the performance of home entry page and lander entry page website

After the test, all traffic volume has been routed to new website page lander

Figure 12: Sessions and Bounced Rate during web traffic rerouting

What is the conversion rate (CVR) for new landing page?

Row	landing_page_url	sessions	orders	conversion_rate
1	/home	2261	72	0.0318
2	/lander-1	2316	94	0.0406

Figure 13: Compare Sessions, Orders, and Conversion rate for landing page test

- The results table displays the conversion rate (CVR) for both landing pages. The conversion difference between /home and /lander-1 = $0.0406 - 0.0318 = 0.0088$. That means, Lander-1 has a slightly higher CVR, generating approximately 0.0088 more orders per session than the home page.
- We had 22972 sessions since after rerouting all traffic to lander-1. So we got $22972 \text{ (sessions)} * 0.0088 \text{ (incremental conversion rate)} = 202 \text{ (incremental orders)}$ since the home page A/B test concluded.
- If we talk about it in terms of orders per month, this is generating an extra 50 orders per month for roughly 4 months (7/29 ~ 11/27).

Showcase full funnel traffic volume from each of the two pages to orders.

Row	from_landing_page	sessions	to_product	to_mrfuzzy	to_cart	to_shipping	to_billing	to_thankyou	product_to_thank_you_CVR
1	homepage	2261	942	684	296	200	168	72	0.0764
2	custom_lender	2316	1083	772	348	231	197	94	0.0868

Figure 14: Full funnel analysis on traffic volume for landing page test

- The table shows the total web traffic for the whole consumer journey click through from landing on the homepage, viewing the product details, adding to cart, to filling in shipping and billing information, and finally complete the order by checking out.
- The custom lander page has a conversation rate of 8.68% whereas for the homepage it is 7.64%. Customers seem to be liking the new lander page and are more likely to purchase product when directed through the new lander page.

Showcase full funnel conversion from each of the two pages to orders.

Row	from_landing_page	lander_to_product	product_to_mrfuzzy	mrfuzzy_to_cart	cart_to_shipping	shipping_to_billing	billing_to_checkout
1	homepage	0.4166	0.7261	0.4327	0.6757	0.84	0.4286
2	custom_lender	0.4676	0.7128	0.4508	0.6638	0.8528	0.4772

Figure 15: Full funnel analysis on conversions for landing page test

- The final table reveals what percentage of people click through from the lander, products, ... etc.
- This new analysis shows that the new landing page custom /lander-1 has a click-through rate that is 5.1% higher than the previous landing page /home and comparable rates throughout the rest of the conversion funnel.

The website manager ran a new custom billing page (/billing-2) in a 50/50 A/B test against the original billing page(/billing) from 2012, Sep 09 — 2013, Jan 05.

Evaluate the new billing page performance by comparing conversion rates between two groups.

Row	from_billing_page	billing_sessions	billing_to_thankyou	billing_to_thankyou_CVR
1	/billing	1656	750	0.4529
2	/billing-2	1650	1024	0.6206

Figure 16: Compare Billing to Thank you page conversion for landing page test

- The results table displays the conversion rate (CVR) for both billing pages. New billing page (/billing-2) has a higher conversion rate which means users are more likely to check out their payments.
- The conversion lift is $0.6206 - 0.4529 = 0.1677$. That means, /billing-2 leads approximately 0.1677 more checkout orders per session than the original billing page.
- We had 2963 billing sessions since after rerouting all traffic to billing-2. So we got $2963 \text{ (billing sessions)} * 0.1677 \text{ (incremental conversion rate)} = 497 \text{ (incremental orders)}$ since the home page A/B test concluded.
- If we talk about it in terms of orders per month, this is generating an extra 124 orders per month for roughly 4 months (01/06/2013 ~ 05/06/2013).

Weekly trend of traffic volume through /billing and /billing-2.

Row	week_start_at	to_thankyou_CVR	billing_sessions	new_billing_sessions
3	2012-08-12	0.3861	101	0
4	2012-08-19	0.4044	136	0
5	2012-08-26	0.437	119	0
6	2012-09-02	0.4828	116	0
7	2012-09-09	0.6	62	58
8	2012-09-16	0.5067	70	80
9	2012-09-23	0.5263	75	58
10	2012-09-30	0.5403	63	61
11	2012-10-07	0.5407	60	75
12	2012-10-14	0.5776	83	78
13	2012-10-21	0.5307	84	95
14	2012-10-28	0.5093	89	72
15	2012-11-04	0.5449	84	83
16	2012-11-11	0.526	102	90
17	2012-11-18	0.5198	208	221
18	2012-11-25	0.5203	176	168
19	2012-12-02	0.5731	130	123
20	2012-12-09	0.5083	129	113
21	2012-12-16	0.5212	120	139
22	2012-12-23	0.5522	64	70
23	2012-12-30	0.5676	19	18
24	2013-01-01	0.5872	53	56
25	2013-01-06	0.6642	0	137
26	2013-01-13	0.6603	0	156

Before test, the conversion of original billing page is around 4.4%

Billing Page test is being conducted to judge the performance of check out conversion

After the test, all traffic volume has been routed to new billing page, the conversion of new billing page is over 6.6%, which is a nice increase.

Figure 17: Weekly Trend of traffic volume for billing page test

What is the revenue per billing sessions for the new launched billing page?

Row	pageview_url ▼	sessions ▼	revenue_per_session	margin_per_session
1	/billing	1656	22.6404	13.8134
2	/billing-2	1650	31.0241	18.9285

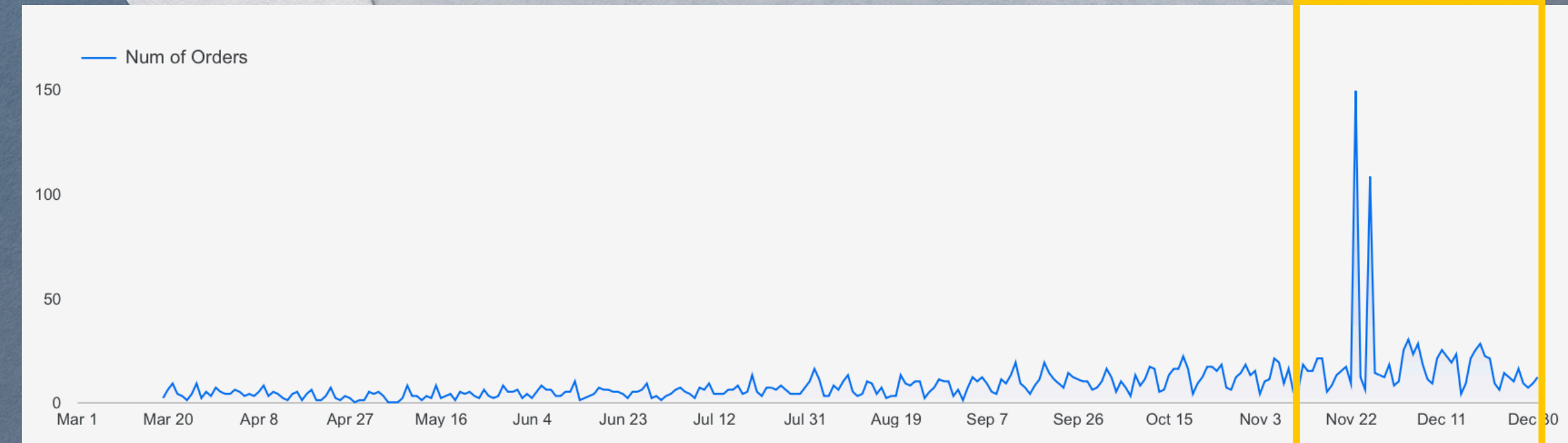
Figure 18: Compare Revenue per session, Margin per session for billing page test

- \$22.64 in revenue was generated per billing page session for the existing billing page (/billing)
- \$31.02 in revenue per billing page session for new custom billing page (/billing-2)
- Lift: \$8.38 per billing page view. This represents a significant increase in revenue of \$8.38 per billing session — Every time a customer sees the billing page, we are now making \$8.38 more than we were previously.
- We got 604 billing sessions for the past month with a lift of \$8.38 per billing session. Therefore, the extra estimated monthly impact of the new billing page is $604 * \$8.38 = \5061.52 over the past month.

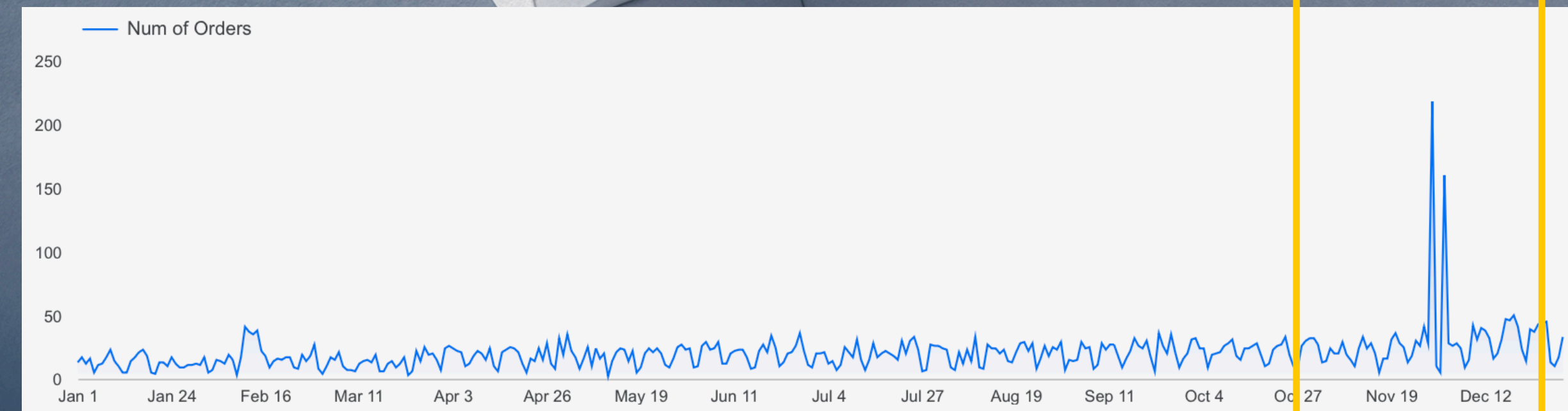
Section 3 - Business patterns and analyzing seasonality

Peak Holiday Seasonality

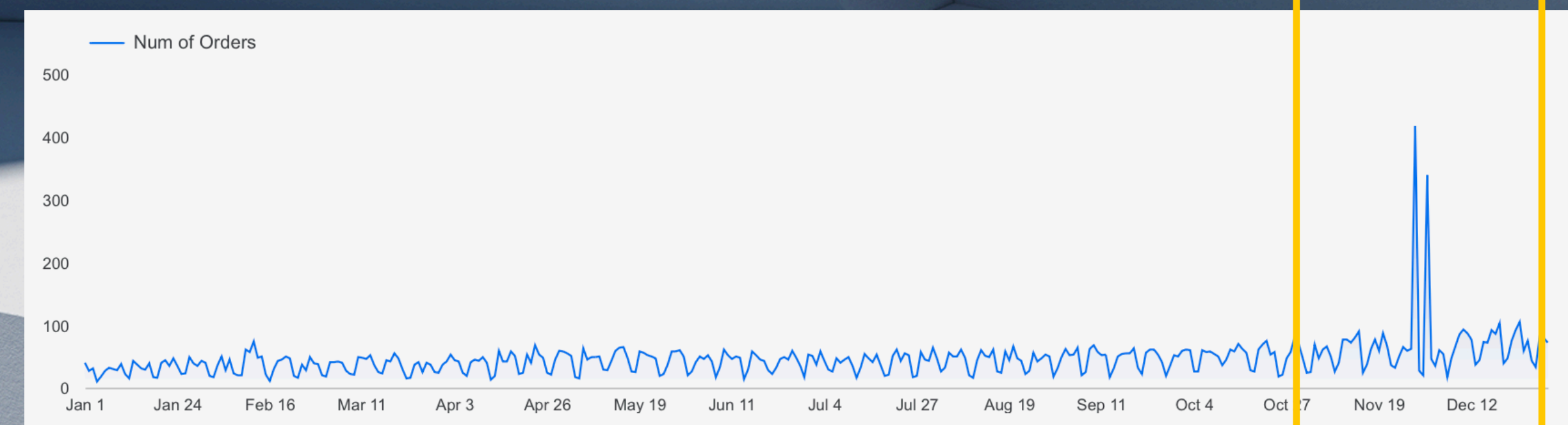
- Compared trend of orders sold over past 3 years, sales is highly affected by the holiday effects like Black Friday and Cyber Monday.
- We see a high demand on our products from potentially those consumers who wanna to send a year-end gift. Most likely couples, teachers, and parents would be our marketing campaigns' target consumer groups.



Number of Orders in Year 2012



Number of Orders in Year 2013



Number of Orders in Year 2014

Dive deep into the holiday effect on orders

Row	year ▼	holiday_avg_orders	nonHoliday_avg_orders ▼	rate ▼
1	2012	129.0	8.0	16.0
2	2013	130.0	20.0	7.0
3	2014	378.0	44.0	9.0

Figure 19: Average order volume between holiday and non-holiday by year

- The table displays the average number of orders between non-holiday vs holiday days for the past 3 years, as well as the weight.
- Holiday (i.e. Black Friday and Cyber Monday) brings much sales for the business. It was about 9:1 ratio comparing holiday season and non-holiday season.
- This is a good sign for the business to optimize the inventory stocks during these holiday seasons and prepare early for consumer service support.

Monday & Friday Effect



Figure 20: Order volume between weekends vs. weekdays

- An interesting trend shows people would more likely to purchase on Monday and Friday.
- Weekend usually have less orders than weekdays.

End